TEXORA · LECTURE SHEET

Historical Development of the Subject and Collection of Data

What statistics means and where it came from, what it can and cannot do, and how reliable data are collected — primary and secondary sources, census and sample inquiry, and the art of the questionnaire.

STAT-2109 · PROBABILITY AND STATISTICS · TOPIC 1 · LEVEL 2 TERM 1

1. Meaning and scope of statistics

Statistics is the science of collecting, classifying, presenting, analysing and interpreting numerical data so that sound decisions can be taken under uncertainty. In the plural sense the word simply means numerical facts — population figures, production records, examination marks. In the singular sense it means the body of methods used to handle such facts. A single isolated number is not statistics; statistics deals with aggregates of facts, affected by many causes, expressed numerically and collected with reasonable accuracy for a stated purpose.

The scope of the subject is now very wide. Wherever numerical evidence matters, statistical method follows: economics and business, government planning, industry and quality control, agriculture, medicine and biological research, education, and engineering. In textile engineering, for example, fibre and yarn test results, fabric inspection counts and machine efficiency records are all statistical data, and no mill can control quality without them. This course develops the core ideas — description, averages, dispersion, probability and inference — on which all of those applications rest.

2. A brief history of the subject

The word itself records the origin. It derives from the Latin *status*, the Italian *statista* (a statesman) and the German *Statistik* — in every case, the facts a state needs about itself: how many people, how much land, how much revenue. Long before the word existed, the practice did. Ancient censuses were taken in Egypt for taxation and labour for building, and in Rome the census of citizens and property was a regular institution. Counting people and wealth is the oldest branch of statistics.

PERIOD / PERSON	CONTRIBUTION
Ancient world	Censuses in Egypt and Rome; records of population, land and taxes kept by the state.

John Graunt, 1662	*Natural and Political Observations on the Bills of Mortality* — the first systematic analysis of birth and death records in London; founder of demography and vital statistics.
Pascal and Fermat, 1654	Correspondence on problems of gambling founded the mathematical theory of probability, the second pillar of the subject.
Gottfried Achenwall, 1749	Used the word *Statistik* in its modern academic sense at Göttingen; honoured as the **father of statistics**.
Gauss and Laplace	The normal curve of errors and the method of least squares gave measurement science its statistical foundation.
Karl Pearson	Founded the first statistics department (University College London, 1911), the journal *Biometrika*, correlation and the chi-square test.
R. A. Fisher	Built modern inference: analysis of variance, design of experiments, maximum likelihood — the working toolkit of twentieth-century research.

Two streams therefore meet in the modern subject: the *descriptive* stream, from state records and Graunt's tables, concerned with summarising data; and the *probability* stream, from Pascal and Fermat onward, concerned with reasoning from a sample to the whole. Every later topic in this course belongs to one stream or the other.

3. Functions and importance of statistics

- **It simplifies complexity.** A mass of raw figures is reduced to a few meaningful summaries — totals, averages, percentages — that the mind can grasp.
- **It presents facts in definite form.** Statements supported by numbers carry more weight than vague impressions: "wastage rose from 4% to 7%" is a fact; "wastage rose a lot" is an opinion.
- **It aids planning.** Governments plan food, housing, education and health services from population and production statistics; no modern plan is possible without them.
- **It serves economics and business.** Demand forecasting, cost analysis, market research and index numbers all rest on statistical method.
- **It controls quality in industry.** Control charts and sampling inspection keep a process within limits and detect faults early — central to textile production, where yarn strength and fabric defects are judged statistically.
- **It makes research possible.** Designed experiments and tests of significance let a researcher separate a real effect from chance variation.

- **It supports comparison.** Ratios, percentages and averages place different places, periods and groups on a common footing.

4. Limitations of statistics

Statistics is a powerful tool, but it has firm limits, and a good user states them plainly. First, it deals only with facts that can be expressed numerically; qualities such as honesty, efficiency or beauty enter only through some numerical proxy. Second, it studies aggregates, not individuals — an average income says nothing about any one person. Third, its results are true only on average and in the long run. Fourth, statistics can be misused: in untrained or dishonest hands, selective figures can be made to prove almost anything. Fifth, the conclusions are only as good as the data — faulty collection gives faulty results, however elegant the later analysis. The remedy is not to distrust statistics, but to insist on honest data and competent method.

5. Data: primary and secondary

Primary data are collected first-hand by the investigator (or an agency) for the specific inquiry in hand. **Secondary data** already exist, collected by someone else for another purpose and later reused. The distinction is one of use, not of the numbers themselves: census figures are primary for the census office and secondary for everyone else.

POINT	PRIMARY DATA	SECONDARY DATA
Originality	Original, collected for this inquiry	Already collected for another purpose
Cost and time	Costly and slow to collect	Cheap and quickly available
Suitability	Fits the purpose exactly	May need adjustment of units, definitions or coverage
Control	Investigator controls definitions and quality	Quality and method of the original collection must be checked
Examples	A mill survey of loom efficiency carried out this term	Published census reports, trade journals, official yearbooks

Before using secondary data, check the source and its reputation, the definitions and units used, the period and area covered, and whether the sample or coverage suits your purpose. Data copied uncritically from an unsuitable source are a common cause of worthless conclusions.

6. Methods of collecting primary data

METHOD	HOW IT WORKS	MERITS AND DEMERITS
Direct personal interview	The investigator meets respondents face to face and asks the questions.	Accurate and complete answers; doubts can be cleared. Costly, slow, and limited to a small area.
Indirect oral investigation	Information is taken from witnesses or third parties who know the facts, not from the persons directly concerned.	Useful when direct contact is impossible or the topic is sensitive. Answers may be coloured by bias.
Mailed questionnaire	A printed or online form is sent to respondents, who fill it in and return it.	Cheapest for a wide, scattered area. Low response rate, and unusable where respondents cannot read or write well.
Schedules through enumerators	Trained enumerators visit respondents, ask the questions and fill in a standard schedule.	High response and reliable entries; used in censuses. Most expensive method; needs trained staff.
Observation	The investigator records what actually happens — counts, timings, defects — rather than asking.	Objective and free of reporting bias. Slow, and some facts cannot be observed at all.

The choice among these is a balance of accuracy, cost, time and the nature of the respondents. A small, careful inquiry often uses personal interview or observation; a national inquiry must use enumerators or mailed forms.

7. The questionnaire: qualities of a good one

Where a questionnaire or schedule is used, its quality decides the quality of the data. A good questionnaire has few questions, each short, simple and unambiguous; questions follow a logical order, moving from general to specific; leading questions that suggest the answer are avoided; personal or embarrassing questions are kept to the end, or omitted; clear instructions for filling in are printed with it; and, where possible, it is tested on a small pilot group before full use. A question that half the respondents understand differently is worse than no question, because its answers cannot be interpreted at all.

8. Census versus sample inquiry

A **census** (complete enumeration) collects information from every unit of the population. A **sample inquiry** studies a carefully chosen part and generalises to the whole. Sampling is not a poor substitute for a census; properly done, it is often *more* accurate, because resources can be concentrated on careful measurement of fewer units.

POINT	CENSUS INQUIRY	SAMPLE INQUIRY
Coverage	Every unit of the population	A representative part only
Cost and time	Very high cost; long time to complete and process	Economical and quick
Accuracy	No sampling error, but large-scale work invites careless errors	Small, measurable sampling error; closer supervision possible
Destructive testing	Impossible (cannot break every yarn or bulb to test it)	The only possible method
Use	Population census, complete stock records	Quality control, market surveys, opinion polls, most research

9. Sources of secondary data and first steps with data

The main sources of secondary data are official publications of governments and statistical offices (censuses, surveys, yearbooks), reports of international bodies such as the United Nations and the World Bank, publications of trade and industry associations, research journals and theses, and newspapers and commercial databases. Whatever the source, the critical checks of Section 5 apply before a single figure is used.

Once collected, raw data are almost unusable in their original form. The first steps are **classification** — grouping observations by a common characteristic (by class, place, time or quality) — and **tabulation** — arranging the classified data in rows and columns with titles, units and totals stated. Good classification is exhaustive, mutually exclusive, stable over the inquiry and suited to the purpose. Topic 2 takes the next step: building a frequency distribution from classified data and presenting it in tables and diagrams.

WORKED EXAMPLE

In a class survey, 24 of 45 students reported that they travel to the institute by bus. The percentage using the bus is

$$24/45 \times 100 = 53.3\%$$

A count (24) has become a comparable figure (53.3%) — the simplest statistical summary, and the form in which the result would be reported and compared with other classes or other terms.

Glossary

TERM	MEANING
Statistics	The science of collecting, classifying, analysing and interpreting numerical data for decision making.
Population	The complete set of units about which information is required.
Sample	A part of the population selected for study, from which conclusions about the whole are drawn.
Primary data	Data collected first-hand by the investigator for the inquiry in hand.
Secondary data	Data already collected by another agency for another purpose, reused by the investigator.
Census	An inquiry covering every unit of the population (complete enumeration).
Questionnaire	A set of written questions sent to respondents, who fill in the answers themselves.
Schedule	A form of questions filled in by a trained enumerator during a visit to the respondent.
Enumerator	A trained field worker who visits respondents and records their answers on a schedule.
Classification	Grouping data by a common characteristic as the first step in organising them.

Key points & summary

- Statistics began as the facts of the state — the word traces to Latin *status*, Italian *statista* and German *Statistik*; Achenwall (1749) is called the father of statistics.
- Graunt's mortality tables (1662) founded descriptive statistics; Pascal and Fermat (1654) founded probability; Gauss gave the normal curve; Pearson and Fisher built the modern subject.
- Statistics simplifies, presents, plans, compares and controls quality — but it handles only numerical aggregates, is true on average, and is only as good as its data.
- Primary data are collected first-hand (interview, questionnaire, enumerators, observation); secondary data are reused and must be checked for source, definitions and suitability.
- A sample inquiry can beat a census on cost, speed and even accuracy, and is the only option for destructive testing.
- A good questionnaire is short, clear, logically ordered, free of leading questions and pilot tested.
- Classification and tabulation turn collected figures into usable data — the bridge to frequency distributions.

Practice

Answer in your own words; brief model answers follow.

1. Explain, with the origin of the word, why statistics was first a concern of the state.
2. State two functions of statistics in industry and two limitations of the subject.
3. A researcher takes last year's published production figures for a new study. Are these primary or secondary data for her? Justify your answer and list two checks she should make.
4. Choose a collection method for surveying the reading habits of students spread across the whole country, and defend your choice against one alternative.
5. Why can a sample inquiry be more accurate than a census, even though it covers fewer units?
6. Rewrite this faulty questionnaire item into a good one: "Do you not agree that the library is excellent and badly in need of more books?"

Answer key

1. The word comes from Latin *status* / German *Statistik* — the science of the state. Rulers needed counts of people, land and revenue for taxation, armies and administration, so the earliest statistics (Egyptian and Roman censuses) were state records.
2. Industry: quality control through sampling inspection and control charts; production planning and forecasting from past records. Limitations (any two): it treats only numerical facts; it describes aggregates, not individuals; its results hold on average, not in every case; it can be misused with selective figures.
3. Secondary data — they were collected by another agency for another purpose. Checks (any two): reputation of the source; definitions and units used; period and area covered against her needs.
4. Mailed (or online) questionnaire: cheap and able to reach a scattered national population quickly. Against enumerators: visiting every student nationally would be far too costly and slow, though it would give a higher response rate.
5. Because effort and money can be concentrated on fewer units: better training, closer supervision and more careful measurement reduce non-sampling errors, which in a huge census often exceed the small, measurable sampling error.
6. Split it and remove the leading wording, e.g. two neutral items: "How would you rate the library collection: very good / good / fair / poor?" and "Does the library need more books: yes / no?"